

Question: How Can We Plot a 3D Surface?

Step-by-Step Sketching Method

To draw a surface $z = f(x, y)$, we reconstruct it from 2D slices:

- 1 **Step 1 (2D Level Curves):** On the xy -plane, draw curves for multiple constant heights $f(x, y) = k$.
- 2 **Step 2 (Lift to 3D):** In $Oxyz$ space, lift each 2D curve up to its actual height $z = k$.
- 3 **Step 3 (Form 3D Surface):** Connect/loft the curves together with a smooth surface mesh.



Step 1: Draw 2D $\rightarrow$ Step 2: Lift to $z = k \rightarrow$ Step 3: Form Surface